Topic 3: Principal Component Analysis (PCA)

Question 1:

Consider the following dataset:

```
X = [[2, 4],
   [3, 5],
   [4, 6],
   [5, 8],
   [6, 9]]
```

Write a Python program to perform the basic steps of PCA.

1. Convert the dataset into a NumPy array.
2. Calculate the mean of each feature.
3. Center the dataset by subtracting the feature means, then display it.

Question 2:

Given the dataset

```
X = [[2, 4],
   [3, 5],
   [4, 6],
   [5, 8],
   [6, 9]]
```

Write a Python program to:

1. Center the data and compute the covariance matrix.
2. Find the eigenvalues and eigenvectors of the covariance matrix.
3. Sort the eigenvalues in descending order.
4. Identify the first principal component.

Question 3:

Consider the dataset:

X = [[2, 4, 1],
[3, 5, 2],
[4, 6, 3],
[5, 8, 4],
[6, 9, 5]]

Perform PCA using Python.

1. Center the dataset and compute the covariance matrix.
2. Find its eigenvalues and eigenvectors.
3. Sort the principal components according to decreasing eigenvalues.
4. Calculate the percentage of variance explained by each component.
5. Project the centered dataset onto the first two principal components.